

CHANGELOG

Changelog

All notable changes to this project will be documented in this file.

The format is based on Keep a Changelog, and this project adheres to Semantic Versioning.

[Unreleased]

Added

- **Wave 88 — buf-lint clean + Production Readiness Review (HXC-1628 done; HXC-1286 executed, stays open):**
 - `api/v1/buf.yaml` (HXC-1628) — `buf lint` now exits clean: 5 STYLE/naming rules excepted with per-rule rationale (proto-at-module-root layout; domain-named RPC request/response types incl. the shared `KernelRequest`; `GPUProvider` service name), `DEFAULT`->`STANDARD` category. `buf build` still passes; removing the except block re-fires 25 violations.
 - `docs/PRODUCTION_READINESS_REVIEW.md` (HXC-1286) — an 80-item production-readiness checklist with an HONEST assessment: **60/80 PASS (75%)**, below the 95% closure bar, so the ticket stays Queued with its gap-list (re-enable CI gates, macOS wireguard-go userspace, govulncheck/SBOM, Makefile migrate wiring, mTLS e2e evidence, darwin GPU probes, covgate enforcement, HelixQA challenge evidence). PASS rows cite real files/tests; NOT-READY rows cite the specific gap. `docs_chain`-tracked.
- **Wave 87 — api/v1 proto schema extensions (HXC-1293/1165/1294/1295/1187/1527; buf-regenerated, conductor-gated, backward-compatible):**
 - `api/v1` — Node +tier/trust_level/compute_class/arch + nested `NodeThermal` (HXC-1293, absorbing HXC-1165 `trust_level+thermal`); `NodeResources` +`npu_tops/fpga_logic_elements/tflops_gpu` (HXC-1294); `ScheduleJobRequest` +`allowed_tiers/min_tier/power_budget_watts` (HXC-1295); new `edge.proto` `EdgeWorkUnit/EdgeWorkResult` (HXC-1187); new `gpu_proxy.proto` `GPUProvider gRPC` service (`DispatchKernel` + `DispatchKernelStream`, `KernelRequest/Response/Chunk`) (HXC-1527). All new field numbers appended (nothing renumbered/removed); 9 round-trip tests prove field preservation; field-tag mutations fail them; `buf` generate idempotent; whole-tree build + consumer cmd tests green.

- **Wave 86 — proto service stubs build + generate (HXC-1117; buf now installed; conductor-gated):**
 - `api/v1` — the proto module was broken (`buf.yaml` mis-placed at `api/` so bare import `"types.proto"` could not resolve; `api/buf.gen.yaml` malformed for `buf v1.70`). Relocated the `buf` module root to `api/v1/` (matching `api/v1/go.mod`), removed the malformed `gen` config, and regenerated all 7 services' Go stubs (`node/session/scheduler/health/build/advisory/security`) with `protoc-gen-go/-go-grpc`. Idempotent; package stays `helixv1`. `api/v1` module + whole-tree build + consumer cmd tests green; `TestCreateSessionRequest_RoundTrip` proves marshal/unmarshal preserves all fields (mutation corrupting a field tag fails it). `buf-lint` STYLE conformance tracked HXC-1628 (HXC-1117).
- **Wave 85 — streaming SSE E2EE decryption (HXC-1565; security submodule 16ae574; conductor-gated -race + tamper/order/wrong-session mutation):**
 - `security/pkg/e2ee` — `StreamDecryptor` (`e2e_init` + per-chunk) reassembles an ordered sequence of independently-sealed SSE chunk records against the per-request handshake-derived `Session`; any tampered, out-of-order/replayed, or wrong-session chunk POISONS the stream (AEAD tag + counter-nonce ordering), so a broken chunk cannot silently corrupt the reassembled completion. Composed from `Session.Open`; no hand-rolled crypto. Live-relay capture deferred. Sibling HXC-1561 (`ChaCha20-Poly1305`) is BLOCKED pending authorization to add `golang.org/x/crypto` to the security module (HXC-1565).
- **Wave 84 — per-request response keypair (HXC-1563; security submodule f09227f; conductor-gated -race + isolation mutation):**
 - `security/pkg/e2ee` — `ResponseKeypair` (fresh ephemeral ML-KEM-768 Initiator per request) + worker `SealResponse` + client `OpenResponse` + `Fingerprint`, composed from existing handshake primitives (no hand-rolled crypto). A response sealed for request-1 is cryptographically unopenable with request-2's key; per-request rotation proven via distinct fingerprints. Mutation reusing one keypair fails the isolation test (HXC-1563).
- **Wave 83 — typed Avro pub/sub for the remaining domain events (HXC-1627; conductor-gated build/vet/-race + REAL-broker integration + load-bearing mutation):**
 - `pkg/events` — `NATSBackend` gains typed `Publish*/Subscribe*` methods for `SessionEvent/SchedulerEvent/AuditEvent` at parity with `NodeEvent`: in `WireAvro` mode each carries an Avro single-object payload (`0xC3 0x01` + fingerprint) decoded via the `SchemaRegistry`; JSON stays default. `TestNATSBackend_TypedAvroWire_RealRoundTrip` (`// go:build integration`) boots a real NATS broker via `brokertest.StartNATS` and asserts the on-wire Avro marker + exact round-trip for all three types. Mutation forcing JSON on `PublishSessionEvent` fails the on-wire assertion. All four domain events now route over the live bus in Avro mode (HXC-1627).
- **Wave 82 — Avro codec wired into the live NATS event bus (HXC-1626; conductor-gated build/vet/-race + REAL-broker integration + load-bearing mutation):**
 - `pkg/events` — `NATSBackend` gains an Avro wire mode: `PublishNodeEvent` emits an Avro single-object payload (`0xC3 0x01` + CRC-64-AVRO fingerprint) when `WireAvro` is selected and `SubscribeNodeEvents` decodes via a `SchemaRegistry` (validated routing); JSON stays the default (back-compat). Concrete Avro schemas +

Encode*/Decode* helpers added for SessionEvent/SchedulerEvent/AuditEvent (backend typed pub/sub for those three tracked HXC-1627). Proven by TestNATSBackend_AvroWire_RealRoundTrip (//go:build integration) which boots a **real NATS+JetStream broker** via `brokertest.StartNATS`, captures the on-wire bytes and asserts the Avro marker (JSON would start with {), and round-trips schema evolution both directions across the live bus. Mutation forcing JSON on the publish path fails the on-wire assertion (HXC-1626).

- **Wave 81 — E2EE framed-transport concurrency + oversize hardening (HXC-1567; conductor-gated -race + load-bearing mutation):**
 - **security submodule (pkg/e2ee, 29a6193)** — the length-prefixed framed Transport (WriteRecord/ReadRecord over `io.ReadWriter`) and its `DefaultMaxRecordSize` guard already existed; added `TestTransportConcurrentRoundTripWithOversizeInjection` closing the closure's composite gap — 16 concurrent independent framed round-trips stay intact while an injected 5 MiB length prefix (cap 1 KiB) is rejected with `ErrRecordTooLarge`, all under `-race`. Mutation dropping the `ReadRecord` oversize guard fails the test (HXC-1567).
- **Wave 80 — Avro event schemas + schema-validated routing (HXC-1119; careful single-threaded TDD build, conductor-gated build/vet/-race + 2 load-bearing mutation bites):**
 - `pkg/events` — a stdlib-only Avro-subset codec: flat primitive records, spec-faithful binary encoding (zig-zag varint int/long, length-prefixed string/bytes, LE-IEEE-754 double), **single-object encoding** framing (`0xC3 0x01` + 8-byte CRC-64-AVRO fingerprint) so each payload is self-describing, a `SchemaRegistry` giving **schema-validated routing** (a payload whose writer fingerprint is unregistered is rejected, never mis-decoded), and **writer→reader schema resolution** for backward+forward-compatible evolution (reader-only fields fill from defaults; writer-only fields skip; a reader-only field without a default is an unresolvable-pair error). `NodeEventSchemaV1→V2` adds a defaulted action field; proven both directions — a V1 consumer reads a V2 payload (skips action) and a V2 consumer reads a V1 payload (synthesizes action from default) — plus concurrent round-trip under `-race`. Live NATS-backend wiring + the other event-type schemas tracked HXC-1626 (HXC-1119).
- **Wave 79 — 2 bounded host-closeable fixes (conductor-gated build/vet/-race + load-bearing mutation bite):**
 - `pkg/multiraft` — the durability persist-error is now observable through a `PUBLIC` seam: a registerable `OnPersistError` hook (+ `LastPersistError` accessor) fires with the shard id + error from the same Ready-loop path that parks the un-persisted Ready and skips Advance (HXC-917); previously the fault was only `log.Printf`'d. Nil hook = prior behavior (HXC-927).
 - `pkg/cloudspot` — the GCP and Azure preemption pollers now honor the injected `WithClock` when computing `PreemptionNotice.Deadline` (both used `time.Now()` directly, silently dropping the clock; only the AWS poller honored it), making all three providers deterministically testable with a fake clock (HXC-1616).
- **Wave 78 — 4 disjoint host-closeable streams + a cross-platform SELinux fix (all conductor-gated build/vet/-race + load-bearing mutation bite, not agent self-reports):**
 - `internal/console` — fixture-driven device-classification hardening: `testdata/ /proc+device-tree` trees for 6 device classes (PS4/PS5/x86-server/RK3588/Raspberry-Pi/generic) with exact-label table assertions;

mutating the PS5 zen-2 SoC-match row flips ps5_zen2 and fails the test. Non-console classes still classify Unknown today (positive labels tracked HXC-1624) (HXC-1158).

- `scripts/cross-compile-agent.sh` + Makefile `cross-agent` target — reproducible ARM64 cross-compilation of `cmd/helix-agent`: emits `linux/arm64` (ELF aarch64), `linux/armv7` (ELF 32-bit ARM EABI5) and `android/arm64` artifacts with a build-UUID via `-ldflags` and a `--version` print path; `crosscompile_test.go` asserts `elf.EM_AARCH64` via `debug/elf`. CGO-enabled cross (no `zig cc/aarch64-gnu-gcc` on host) and on-real-SBC execution are honest SKIPS (HXC-1622/HXC-1623) (HXC-1167).
- `docs/architecture/PHASE_8C_INTEGRATION.md` — code-grounded Phase-8C integration map (attestation→scheduler, e2ee→orchestrator, marketplace) citing real `pkg/resources` symbols; scheduler attestation-wiring labeled PLANNED (verified: no `SetAttested` callers) (HXC-1614).
- `docs/PHASE_8C_EXIT_GATE_EVIDENCE.md` — CLAUDE-1 exit-gate evidence matrix (PROVEN 9 / PARTIAL 6 / NOT-YET 7); PROVEN rows cite real `pkg/metering+pkg/auditproof` tests, NOT-YET rows cite Queued tickets (HXC-1613).
- **containers submodule (pkg/crossbuild)** — SELinux bind-mount relabel made conditional + cross-platform: `:Z private / :z shared` **only when SELinux is active**, empty on macOS / non-SELinux Linux (was hardcoded `:Z`, breaking podman-machine/Docker-Desktop & shared-volume reuse); injectable detection, panic-recovered fail-safe, stress+chaos coverage; CLAUDE-2 §11.4.81 (HXC-1621, submodule 1598f28).
- **Waves 76+77 — two concurrent subagent waves, 8 disjoint host-closeable streams (all independently gated build/vet/-race + load-bearing mutation bite per stream):**
 - `internal/llm` — advisory engine (MIGRATION/SCALING/CONFIG/ALERT/OPTIMIZATION) carrying rationale + confidence + risk, PENDING persistence, a non-LOW-risk approval gate (never auto-applied without explicit approval), and a mandatory LLMsVerifier that rejects hallucinated/unsafe actions; the LLM is an injectable Advisor so the gate is deterministic and host-closeable (HXC-1126).
 - `pkg/benchmark` — micro-benchmarking producing a real, repeatable normalized on-host CPU score (+ typed GPU TFLOPS / NPU TOPS seams), repeatability asserted within a coefficient-of-variation tolerance band (HXC-1308).
 - `pkg/security/e2ee (security submodule)` — gzip-before-encrypt payload compression matching the Chutes envelope framing: measurable wire-size reduction, byte-exact decrypt+gunzip round-trip, only applied when it actually shrinks (HXC-1564).
 - `cmd/dst-sim` — 1000-seed deterministic-simulation gate composing `pkg/porcupine`; exits non-zero naming the offending seed on a linearizability violation (ran 1000 seeds exit 0, deterministic across reruns) (HXC-915).
 - `docs/MVP_ARCHITECTURE.md` — living seven-layer architecture + service-communication Mermaid grounded in real packages, with a Go drift-validator that fails the build if the doc names a path that no longer exists; docs_chain-tracked (HXC-1145).
 - `pkg/gpu` — thread-safe ProviderAdapter registration hooks (Register/List/Get/Lookup) exposing a pool-facing listing, duplicate-name guarded, race-free under `-race` (HXC-1533).
 - `pkg/cloudspot` — real AWS IMDSv2 (token PUT → authenticated GET spot instance-action), Azure scheduled-events, and GCP preemption pollers behind SignalSource, with drain→checkpoint→upload via an injectable

- sink; proven against httptest fixtures replaying the genuine wire protocol (live cloud deferred, HXC-1617) (HXC-1328).
- pkg/tierdetect — real cross-platform host-capability detection (darwin sysctl/arch + linux /dev/kvm,/proc,binfmt behind one build-tagged interface) gating tier requests pre-provision with a typed MissingCapabilityError (HXC-1230).
- **Wave 75 — provider hardening, security & boot readiness (5 items, all independently gated):**
 - pkg/provider/runpod — Provision no longer holds the mutex across the ColdProvision network RPC (concurrent cold provisions overlap; reserved counter preserves capacity), and EndpointOf(id) surfaces the reachable worker endpoint (HXC-919).
 - pkg/marketplaceadapter (**security**) — buildSDL YAML-encodes caller-controlled fields to contain manifest injection, and the Akash reputation gate is no longer bypassable by a zero-value adapter (HXC-918).
 - pkg/provider/chutes — parseRetryAfter HTTP-date branch + past-date clamp now covered by tests (HXC-920).
 - internal/console — BootCoordinator wired into the real Registrar via a ReadyGate so node registration blocks on boot readiness (HXC-1147).
 - docs/guides/phase_02_architecture.md — Phase-02 architecture guide + Mermaid diagram, docs_chain-tracked (HXC-1164).
- **Wave 74 — accelerator probing, power-aware edge & raft durability (5 items, all independently gated build/vet/-race + load-bearing mutation bite):**
 - pkg/resources accelerator reporting: GPUInfo.TFLOPS + Accelerators{NPUTops, FPGALogicElements, QPUPresent}, real per-OS probe (darwin Apple-GPU TFLOPS from system_profiler, linux sysfs PCI catalog) + operator override-label fallback (HXC-1300).
 - pkg/powergater — charging-gated CanAcceptWork() (bool, reason) work-acceptance guard with real darwin pmset / linux sysfs PowerSource (HXC-1175).
 - pkg/edgeheartbeat — battery/charging/thermal/network heartbeat + churn-timeout collector over a real loopback transport, real per-OS TelemetrySource (HXC-1185).
 - pkg/multiraft **durability fix**: a ShardStorage persistence error now **parks** the un-persisted Ready and skips Advance (no false “durably handled”), instead of swallowing the error (HXC-917).
 - docs/NODE_PROVISIONING_BOUNDARY.md — normative no-jailbreak/no-root/no-unlock provisioning boundary, registered in docs_chain (HXC-1146).
- **Foundation distributed-systems library (pkg/)** — a large, growing pure-Go, deterministic, mutation-tested package set. Recent additions include:
 - Consensus/coordination: voting, failconfirm, heartbeatcoalescer, splitbrainalert.
 - Replication/state: mvcc (B-tree time-travel store), antientropy (hinted-handoff + read-repair + Merkle diff), watchmanager (syncd/unsyncd/victim watch groups), crdt/merkle.
 - Scheduling/placement: constraints (Pacemaker 4-type), preempt (value-multiplier), priorityqueue (multifactor aging), workclaim (SKIP-LOCKED), admissioncontrol (N+K reserve), budgetcap, qos, suitability, ewmarank, fallbackchain.
 - GPU/resource & cost: costsched, latencysched, healthmonitor, gpuattest (attestation crypto: challenge/response, proof-of-GPU-work,

O(1) spot-check, device-sealing), attestadmit, quantization, local (TCO), pool, burst.

- Messaging/flow & edge: flowcontrol (K8s APF), idempotent (exactly-once), rebalance (cooperative-sticky), informer (list-watch cache), redundantexec (BOINC trust), edgeregistry, edgeverify, slotmigration, hashslot, healthprobe.
- Testing/simulation: timefault (clock skew/freeze/monotonic injectors), BUGGIFY in testing/dst, phase7matrix (gap-matrix verifier).
- This session (34 pure-Go units):
 - bursthysteresis — BurstController with a MONITOR→SPILL→RECOVER two-threshold hysteresis dead-band (HXC-1504).
 - providerchain — ordered multi-tier provider fallback cascade with retrievable/terminal error classification and injected-clock failover SLA (HXC-1508).
 - gpucatalog — SUPPORTED_GPUS compute-multiplier catalog, LookupMultiplier, and attestation-aware Score (attested above un-attested) (HXC-1592).
 - gravaladmit — HMAC-SHA256 GraVal challenge-response provider admission gate with single-use nonce and graval.verified label (HXC-1523).
 - modelrouter — strategy-to-default-model resolution (latency/throughput/quality/cost) with a typed unknown-strategy error (HXC-1430).
 - gravalverify — GraVal GPU attestation with a VRAM-ratio ≥ 0.95 gate and a concurrent, race-clean BatchVerify pass-rate KPI (HXC-1435, HXC-1436).
 - gepetto — HelixGepetto dual-resource local-vs-Chutes arbitration with a monotonic reserve schedule and a strict >0.80 high-water anti-starvation zero-capacity cut-off (HXC-1446).
 - chutesaccount — Chutes API model-list (TEE/price fields) and /users/me account-balance client over HTTP with Authorization: Bearer (HXC-1429).
 - cmd/burst-controller — utilization-driven binary driving bursthysteresis; emits a “burst allocation request” on SPILL and a RECOVER line, runID-stamped (HXC-1531).
 - tieredcache — hot(memory)/warm(NVMe)/cold(SSD) tiered cache with injected backends + injected clock, idle-demotion + promotion, and per-tier hit stats (HXC-1418).
 - smartrouter — intelligent ModelRouter (latency/throughput/cost/TEE/balanced) that always excludes unhealthy models, with a typed no-eligible-model error (HXC-1539).
 - epochresolve — config-epoch failover slot-ownership conflict resolution (highest-epoch wins, deterministic lexicographic equal-epoch tiebreak, order-independent convergence) (HXC-1397).
 - balancemonitor — USD account-balance monitor with a strict low-balance floor warning, over HTTP with Authorization: Bearer (HXC-1512).
 - scan — Oracle-SCAN-style stable virtual client endpoint: least-loaded healthy-backend routing with a membership-stable name and injected clock (HXC-1403).
 - llmfailover — error-classified LLM failover taxonomy (deterministic per-class fallback paths) plus a sandbox scheduling-hint contract (HXC-1600).

- `llmadapter` — Claude Messages / OpenAI Responses-API request and response shape adapters to OpenAI chat-completions, preserving tool-call order (HXC-1599).
- `deltacrdt` — standalone delta-state CRDTs (G-counter / PN-counter / observed-remove OR-set / LWW-map) with delta-mutators and a convergent (commutative/associative/idempotent) merge (HXC-1409).
- `gputopo` — topology-aware NUMA/NVLink GPU placement scoring that prefers NVLink-connected GPU pairs and falls back to PCIe (HXC-1406).
- `fsresidency` — filesystem residency challenge: per-byte-range ReadAt + SHA-256 verification proving a model file is really present on disk; truncated/missing/mismatch all fail (HXC-1572).
- `economics` — `RewardDistributor` multi-token distribution with treasury/reinvest splits and a conservation invariant (`sum(ledger) + treasury + reinvest == total`), plus `GetParticipantROI` (ROI% + break-even days, typed unknown-participant error) (HXC-1460, HXC-1461).
- `revenueopt` — `RevenueOptimizer.OptimizeAllocation` greedy GPU→marketplace assignment maximising expected revenue, with a TEE→Chutes revenue bias; self-contained types (HXC-1456).
- `exportcontrol` — export-control / country-code KYC gate: a controlled GPU (H100/A100) from a Tier-3 (embargoed) or unknown jurisdiction is rejected with `ErrExportDenied`; Tier-1 is allowed (HXC-1482).
- `devicecatalog` — machine-readable 64-device taxonomy with case-insensitive substring Lookup resolving a discovered CPU model (e.g. Rockchip RK3588) to its tier / trust-level / compute-class (HXC-1331).
- `compliance` — EU AI Act compliance-documentation pipeline: generates a model card + technical-documentation artifact from attestation-log records, with a SHA-256 provenance hash bound to the actual record content (HXC-1481).
- `hybridkex` — hybrid post-quantum key exchange combining X25519 (crypto/ecdh) and ML-KEM-768 (crypto/mlkem) shared secrets via HKDF-SHA256 (group X25519MLKEM768); a tampered ML-KEM ciphertext yields a non-matching key (HXC-1476).
- `carbonsched` — carbon-aware job placement: chooses the lowest-carbon-intensity latency-eligible region (deterministic name tiebreak) with per-job kWh / gCO2 metering (HXC-1480).
- `gpuattest` (additive `multigpu.go`) — multi-GPU node enumeration: an N-GPU node descriptor yields N distinct, independently-verifiable per-device attestation proofs keyed by GPU UUID; rejects duplicate/empty descriptors (HXC-1573).
- `workloadrouter` — `UnifiedManager.RouteWorkload`: concurrent per-marketplace price probes + a weighted composite score (inverse price/latency, health) with a TEE multiplier that re-ranks confidential offers when the workload requires a TEE (HXC-1455).
- `metrics` (additive `tiermetrics.go`) — `TierMetrics` exposes `gpu_tier_utilization`, `gpu_cost_per_hour`, and `provider_health` Prometheus-text series with deterministic ordering and concurrent-safe setters (HXC-1535).
- `cmd/gpu-pool-manager` — GPU pool manager HTTP front-end: `POST /allocate` returns the tier-correct (lowest-tier healthy, name tiebreak) provider as JSON (503 when none healthy) and `GET /`

providers lists providers with health; httpstest-gated, self-contained (HXC-1530).

- marketplaceadapter — MarketplaceAdapter interface (Name/GetCurrentPricing/SubmitWork) with a Name()-keyed dispatch Registry and a ChutesAdapter over HTTP (httpstest fixture, non-2xx rejected) (HXC-1454).
- cmd/e2ee-proxy — transparent ML-KEM-768 (crypto/mlkem) + AES-256-GCM encrypting proxy: EncryptingTransport seals request bodies and DecryptingHandler opens them, so the on-wire payload between proxy and upstream is ciphertext-only while the client receives the correct plaintext; tampered ciphertext is rejected (HXC-1532).
- e2eebench — MeasureHandshakeLatency benchmarks the hybridkex full handshake (median + P95), proving median < 1ms on host and enforcing per-iteration key agreement (ErrKeyDisagreement) (HXC-1556).
- archlint + docs/ARCHITECTURE.md — the hardened L0-L7 architecture diagram and component map, mechanically enforced: archlint parses the component-map table and fails the build if any documented component maps to a package path that does not exist on disk (HXC-1424).
- Wave 67 (5 units, subagent-driven, 5 parallel streams in disjoint packages):
 - multiraft — MultiRaftManager partitioning cluster state into independent per-shard go.etcd.io/raft/v3 groups, each electing its own leader and committing independently via per-shard Propose routing over an in-process RaftTransport; write throughput scales with shard count and a shard re-elects on leader loss while others keep committing (HXC-1387).
 - stonith — STONITH fencing: a FencingAgent interface with IPMI / AWS-EC2 / Azure-ARM / SBD-shared-disk / NoOp drivers and a MultiLevelFencer multi-level fallback, with sink-side fence confirmation; the IPMI real-binary path is an honest SKIP-with-reason when ipmitool is absent (HXC-1401).
 - deviceplugin — Nomad/K8s-style gRPC device-plugin / GRES fingerprinting framework over real gRPC: plugins register full device fingerprints, the registry parses GRES descriptors, atomically applies fingerprints, allocates by request, and rejects oversubscription (HXC-1407).
 - internal/gpu (additive manager_helixpow.go) — dual-workload capacity reservation: ReserveForHelixPoW(fraction) + ChutesCapacity() reporting only the non-reserved remainder, with a strict >0.80 Helix-load Gepetto starvation guard that zeroes Chutes capacity (HXC-1447).
 - chutes (additive config.go) — ChutesMinerConfig and ValidatorConfig deployment descriptors with Validate() invariants (non-empty IDs/hotkeys, GPUCount>0, non-negative cost, positive cache size, TEE consistency) (HXC-1439).
- Wave 68 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 69):
 - dst — standalone FoundationDB-style deterministic simulation harness: single-threaded seeded-RNG event loop with injectable network/disk/logical-clock; same seed => byte-for-byte identical trace + exact failure replay; 1000+ seeded sims (HXC-1419).

- porcupine — self-contained WGL linearizability checker + concurrent history recorder: PASS on linearizable, FAIL (with violating op) on a seeded non-linearizable history (HXC-1421).
- kraft — KRaft-style self-managed Raft metadata quorum (go.etcd.io/raft/v3, no ZooKeeper): replicated CreateTopic/ListTopics + partition-leader assignment, controller election + controller-loss failover (HXC-1417).
- internal/chaos — fault-injection library (PodKill/NetworkPartition/DiskStall/ClockSkew) + canary-guarded runner with automatic rollback on health<SLO; deterministic via injected clock + seeded selection (HXC-1422).
- multiraft (additive lease.go) — LeaseTracker CockroachDB-style leaseholder local reads (no Raft round-trip), follower routing via RaftTransport.SendRead, lease-expiry re-route against an injected clock (HXC-1389).
- Wave 69 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 68):
 - stonith — IPMI credential hardening: BMC password delivered via the IPMI_PASSWORD env + -E (never on the argv / process table) with redacted argv in error strings; tests assert the password appears in neither argv nor errors (HXC-908, closing a wave-67 follow-up).
 - deviceplugin — concurrency test driving ApplyFingerprint/Allocate/Release from 50+ goroutines under -race, proving the Registry never oversubscribes (mutex removal trips the race detector) (HXC-910, closing a wave-67 follow-up).
 - internal/llm — replaced the stub Inference with a real strategy-based router (latency/throughput/quality/cost) over the live model registry, excluding unhealthy/unloaded models with a typed no-eligible-model error and a real injectable Backend seam (HXC-1470).
 - metrics (additive earningsmetrics.go) — EarningsMetrics exposing tao_earnings_total / graval_attestation_status / token_throughput / gpu_utilization Prometheus-text series with deterministic (sorted) ordering + concurrent-safe setters (HXC-1483).
 - internal/gpu (additive attesthook.go) — GraVal attestation hook: AllocateForChutes refuses a GPU that has not passed attestation (ErrAttestationRequired), even on the idempotent re-allocation path; injectable ChutesAttestor seam (HXC-1448).
- Wave 70 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 71):
 - provider/chutes — OpenAI-compatible /v1/chat/completions provider (cpk_Bearer) with HTTP-429 retry+backoff; httptest proves retry-then-success, Bearer header, and typed errors (HXC-1510).
 - pool (additive model.go) — GPUDevice/WorkloadSpec/GPUAllocation/PoolStatus data model with snake_case JSON tags + round-trip/concurrency tests (HXC-1495).
 - quantization (additive awq.go) — AWQ 4-bit default serving format + EstimateVRAM (~25% of fp16) + budget-aware SelectFormat preferring AWQ when fp16 won't fit (HXC-1473).
 - internal/gateway (additive inference.go) — inference route forwarding to an injectable Chutes client; TEERequired => X-E2EE-Enabled routing marker; client failure => 502 (HXC-1469).
 - internal/gpu (additive mig.go) — MIG profile management: standardized vGPU tiers (1g.10gb/2g.20gb/3g.40gb/7g.80gb), in-

memory partitioning with oversubscription rejection + fixed-at-creation immutability (HXC-1449).

- Wave 71 (5 units, subagent-driven, 5 parallel streams in disjoint packages, run concurrently with wave 70; closes 4 prior follow-ups):
 - **multiraft** — RaftTransport async-delivery safety: inbox handler Steps under the shard mutex, outbound messages deferred + flushed after unlock; a new async-transport -race test proves a goroutine-delivering transport is race-free and still commits (HXC-909).
 - **kraft** — CreateTopic returns ErrTopicExists on a conflicting re-create (different partition count); identical re-create stays idempotent (HXC-912).
 - **porcupine** — WGL state-dedup memoization via Model.Equal: verdict-preserving pruning (memo fires on backtracking histories; identical PASS/FAIL + witness vs Equal==nil) (HXC-913).
 - **chutes** — StreamChannel honors ctx cancellation on a blocked read (prompt channel close, no goroutine leak) and closes the owned reader (HXC-911).
 - **marketplaceadapter** (additive akash.go) — Akash (Cosmos/AKT) adapter over an injectable client: reverse-auction pricing, SDL submission, provider-reputation gate, registered in the Name()-dispatch Registry (HXC-1458).
- Wave 72 (5 units, subagent-driven, 5 parallel streams in disjoint packages):
 - **provider/runpod** — RunPodProvider (implements pool.GPUProvider) with a warm-pool fast path over an injectable client (HXC-1515).
 - **provider/aws** — AWSProvider EC2 Spot adapter over an injectable EC2-client interface (no aws-sdk dep): GPU model→instance-type selection (p5/p4de/g5), Spot+tags, TOCTOU-safe capacity gate (HXC-1516).
 - **scheduler** (additive tier_filter.go) — GPURTier-aware filter predicate: hard-exclude disallowed tiers + preference scoring, composes with the existing count filter, fails closed on a nil predicate (HXC-1534).
 - **provider/chutes** — ctx-interruptible 429 backoff (select vs ctx.Done()) + honors Retry-After (delta-seconds/HTTP-date, max(hint,backoff)) (HXC-916, closing a wave-70 follow-up).
 - **grafanadash** (additive tiercost_dash.go) — Phase-8B tier-utilization + cost dashboard generator (panels target gpu_tier_utilization/gpu_cost_per_hour), valid + deterministic (HXC-1551).
- Wave 73 (5 units, subagent-driven, 5 parallel streams in disjoint packages; impl→spec-review→quality-review + independent build/vet/-race/mutation gate):
 - **provider/ionet** — IONetProvider over the io.net REST API (implements pool.GPUProvider, injectable base URL + *http.Client + bearer): DeployCluster returns the backend-minted cluster run-UUID (parsed, never fabricated), HealthCheck derives healthy state, concurrency-safe reserve-under-lock capacity gate; closure proven against a net/http/httptest io.net backend (HXC-1514).
 - **gitops** — ArgoCD ApplicationSet federation client: matrix generator expands one Application per cell, syncPolicy prune+selfHeal, RollingSync canary→tier-2 (50%)→tier-1 (25%) ordering, drift/prune

- detection; logic proven vs an httptest ArgoCD, strictly-live UI capture honestly skipped (follow-up HXC-922) (HXC-1367).
- `internal/federation` — Karmada PropagationPolicy + OverridePolicy engine (CRs modeled as Go structs → YAML, no client-go dep) with constraint-aware (data-locality/latency/cost/compliance) two-level cell selection and failover reselect; deterministic failover proven via in-process FakeHub, live Karmada <60s capture honestly skipped (follow-up HXC-924) (HXC-1364).
- `health` (additive `miner.go`) — `miner-api` + GraVal bootstrap DaemonSet dependency checks wired into the health rollout, naming the specific failing check; `healthy`→`unhealthy` delta proven against an httptest `miner-api` (HXC-1484).
- `discovery` (additive `mdns.go`) — mDNS/DNS-SD advertiser+browser on `_helix-cluster._tcp` (stdlib net + x/net/dnsmesssage): TXT encode/decode + reject-invalid + in-process loopback discovery; real-multicast two-node LAN capture sandbox-skipped (follow-up HXC-925); discovery-only (trust via SPIFFE) (HXC-1347).
- Reconciled stale P0 HXC-904 Phase-4 Build Service: delivered podman-based multi-arch container build orchestration + artifact cache (`internal/build/pkg/build/cmd/helix-build`, green) marked Completed; Bazel REAPI interop split to follow-up HXC-921.
- **Security fix (HIGH):** closed a TOCTOU replay-bypass in `pkg/gpuattest.Verify` — the nonce check and consume are now atomic, with a concurrent-replay -race regression test.
- Every foundation package ships with tests that fail under mutation of the logic they cover (CLAUDE-1 enforcement) and are gated by whole-tree build/vet + -race.
- Initial project scaffold with Go workspace.
- 29 submodules integrated at project root.
- Directory structure: `cmd/`, `pkg/`, `internal/`, `api/`, `web/`, `scripts/`, `deploy/`, `test/`.

Changed

- **Governance:** CLAUDE-3 / AGENT-2 / QWEN-2 documentation continuous-sync guarantee added (restates and cites Constitution §11.4.106 docs-chain); `.docs_chain/contexts/tracked_docs.yaml` extended to enforce README, CLAUDE/AGENTS/QWEN, CHANGELOG, FOUNDATION_PACKAGES, and CODEGRAPH export-sync.

[0.1.0-dev] - 2026-05-30

Added

- Phase 0 bootstrap: submodule cloning and workspace initialization.